PROJECT PROPOSAL
MLinAPP

1. Project Title, Project Number and Project Owner

Detection of Anomalous Behaviour in Industrial Robot, 2026/AM01, Alessio Mascolini

2. Project Team

Elziario Paduano (348055),
Marco Pavanati (341333),
Klejsi Zeqaj (344184),

3. Project Background

The project came out from the need to advance predictive maintenance for Kuka industrial robots, which frequently experience subtle performance degradation, such as reduced speed and precision. To address this, we are analyzing multivariate time-series data collected directly from the robot's sensors, including joint angles, velocity, current, and power usage. Our primary motivation is to investigate whether implementing an Adversarial Autoencoder (AAE) offers a tangible improvement over traditional autoencoders in detecting these minor deviations from normal operational behavior. The main challenges involve handling the high dimensionality of the continuous sensor data and facing the severe class imbalance inherent to anomaly detection, making the theoretical selection of robust evaluation metrics a critical hurdle.

4. Project Scope

The scope of this project involves the offline development and evaluation of an Adversarial Autoencoder (AAE) alongside a baseline standard Autoencoder, utilizing pre-recorded multivariate time-series data from Kuka industrial robots. The core workflow involves preprocessing the continuous sensor data, implementing both network architectures, and conducting a comparative analysis driven by appropriate evaluation metrics tailored for highly imbalanced anomaly detection. The project deliberately excludes real-time deployment on factory networks, physical hardware interventions, root-cause mechanical diagnosis, and cross-machine generalization. Ultimately, the result will be a documented codebase paired with a report that quantifies whether the adversarial component improves the detection of subtle operational degradation.

5. High-Level Requirements

The only requirements needed are software-based. First, employ an Adversarial Autoencoder based architecture and second evaluate it with appropriate metrics.

6. Implementation Plan

Our implementation plan begins with a comprehensive literature review focusing on deep-learning-based anomaly detection architectures and mathematical frameworks for evaluating highly imbalanced time-series data. Next, we will conduct an exploratory dataset analysis on the Kuka robot's sensor outputs to identify noise, correlations, and temporal dependencies, guiding our choice of preprocessing techniques. We will then implement a standard Autoencoder (AE) to serve as our baseline model, followed by the development of the Adversarial Autoencoder (AAE). Subsequentially, we will define a robust suite of evaluation metrics that are mathematically justified for class-imbalanced datasets. Finally, we will execute comparative experiments to analyze the reconstruction error distributions of both models, concluding with a quality assessment to quantify exactly how much the adversarial component enhances anomaly detection performance.

7. Tentative Timeline/Schedule

Week	*Project Phase*	*Key Activities*	*Target Outcomes*
Week 1	*Research & Preprocessing*	*Research AE/AAE anomaly detection and evaluation metrics. Pre-load dataset tensors and implement preprocessing.*	*Defined architectural references; finalized data preprocessing pipeline.*
Week 2	*Baseline AE & Metrics*	*Implement and train the standard Autoencoder. Formalize the mathematical evaluation framework.*	*Trained baseline model; defined evaluation suite.*
Week 3	*AAE Implementation*	*Develop the Adversarial Autoencoder architecture.*	*Trained AAE model.*
Week 4	*Optimization & Thresholding*	*Hyperparameter-tune both models. Compute the anomaly detection threshold.*	*Optimized models; operational threshold*
Week 5	*Comparison & Assessment*	*Execute comparative experiments. Analyze error distributions to quantify AAE vs. AE performance differences and compile findings.*	*Comprehensive academic report; documented codebase; final quality assessment.*

Expected delivery date 13th July

8. Confidentiality Agreement

*By participating in this project, each group member agrees that all shared data, developed code, reports, and related documentation are confidential. These materials may be used solely for the purpose of completing **this course exam** and for no other purpose.*
Any associated repositories or materials must not be made public or shared outside the group without prior consent.

APPROVAL TO PROCEED

We approve the project as described above, and authorize the team to proceed.

Name	Date